

Glens of Tekoa Field Study

North Canterbury, New Zealand

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Abstract

During my semester abroad in New Zealand in the Spring of 2017, I spent a week in mid-April studying igneous rock units at the Glens of Tekoa - an area in North Canterbury. During this time I collected field data in the form of notes, photographs, sketches, and hand samples. I also drew a map, which illustrates the location of each rock unit. The objective of this work was to compare the composition of an agglomerate outcrop with the composition of the surrounding lithology, syenite. A study of the rocks in this area was published in 1991 by S. Weaver and R. Pankhurst, providing a precise Rb-Sr for the Mandamus Igneous Complex, the complex that my area of investigation is a part of. This paper was relied upon during my fieldwork, but does not mention the outcrop I studied, which is clearly agglomerate, and presented on the Weaver-Pankhurst map as syenite. The point of my work was to improve the classification of outcrops in the area, and to speculate on the possible origins of the interesting outcrop.

Introduction

In April of 2017, I spent a week investigating the physical structure and composition of an agglomerate outcrop (and surrounding areas), on the side of a hill at the Glens of Tekoa in North Canterbury, New Zealand. The lithology of this hill was previously classified as entirely syenite, in the most recent academic study done by S. Weaver and R. Pankhurst in 1991. My findings suggest that the area contains more than syenite, following from the observation of an agglomerate outcrop in the area described in Weaver and Pankhurst's paper. Data was collected in the field and analyzed in the lab. Differences in K abundance, along with significant variation in minor element compositions among the outcrops investigated, suggest a short and explosive history of formation. Hydrothermal alteration and differential cooling are speculated to be contributors to elemental variation.

Methods and Materials

This project relied on typical geologists tools: a rock hammer, sample bags and labels, notebook, hand lens, map board and base map, compass, and a bottle of HCl. These materials were used in the field to record locations and data from samples taken at each location. Lab analysis relied on a rock saw to expose fresh faces on hand samples, and a portable XRF for geochemical analysis of element composition. *(Note: My original samples were lost after I left the country, during the move-in to the new geology building. My professor was kind enough to go back to the field and re-collect hand samples, and send me pXRF data via email.)*

Fieldwork and Lab Analysis

I worked independently from sunrise to sunset each day, taking hand samples, photographs, sketches, and notes on structure, composition, and precise location from each significant outcrop visited. I began at an interesting outcrop (Figure 3a - GT1), in situ and of apparently agglomerate composition. While searching around for similar outcrops, I observed two smaller boulders at the foot of the hill beneath this outcrop, likely sourced from GT1. No other outcrops of similar composition were found in this area.

I spent the evenings organizing data. After field work was completed, I further organized the data and reviewed outcrop descriptions based on a more detailed inspection of hand samples obtained from each. Geochemical analysis was done with a portable XRF to investigate mineral composition; data is presented in the form of relative mineral abundance between various outcrops, displayed in the form of spider diagrams, and a bar graph. Relative abundance of major elements is also shown as a gradient from peak to valley, on the hill I investigated.

Data and Results

The following graphs show a comparison between elemental abundance in matrix vs. clasts of the agglomerate outcrop, GT1. Graphs are provided for both major and minor elements of GT1. GT1 is also plotted against pXRF data from an adjacent syenite outcrop for local lithological comparison. Major element concentrations for each outcrop are also plotted in order of decreasing altitude. A map of the field site, outcrop descriptions from field notes, and images of important features are included.

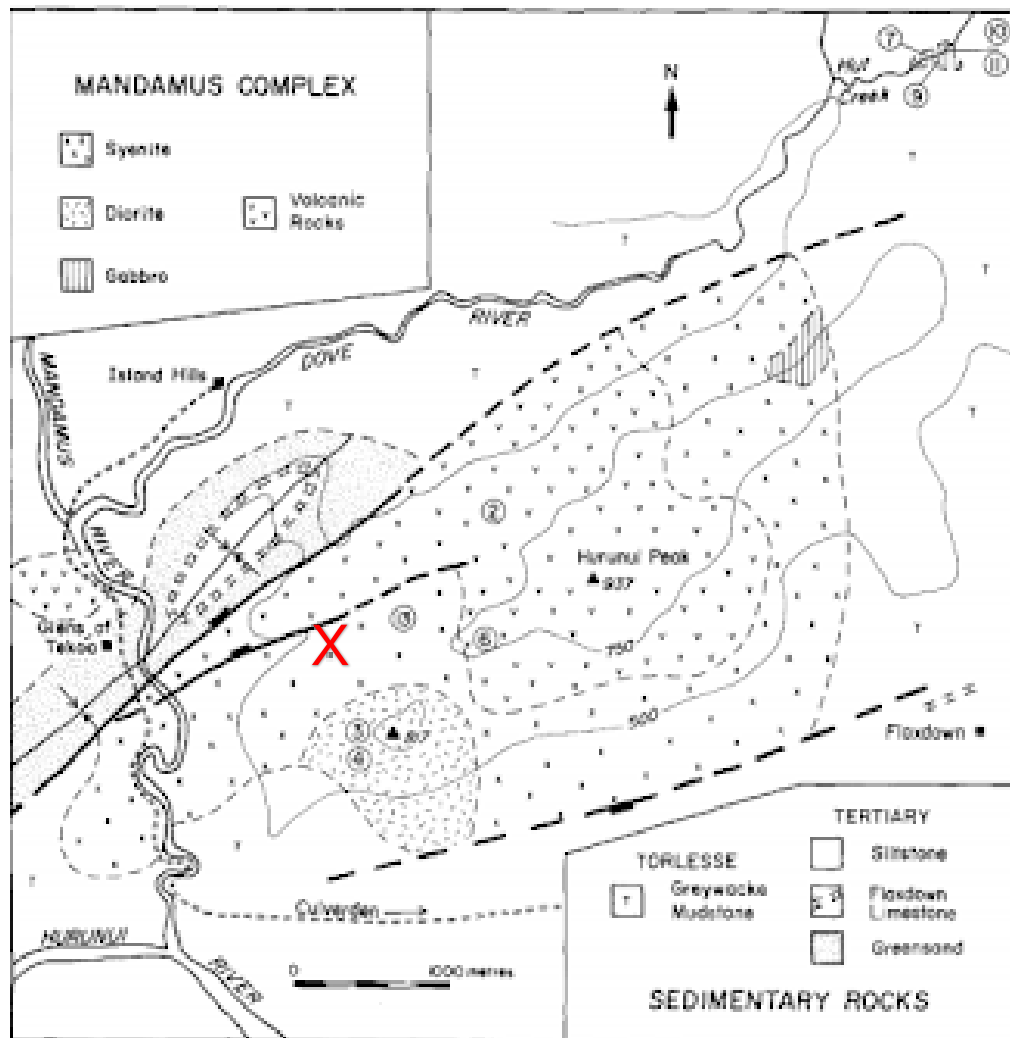


Figure 1: Map from Weaver and Pankhurst - the area investigated in my project is marked by the red X.

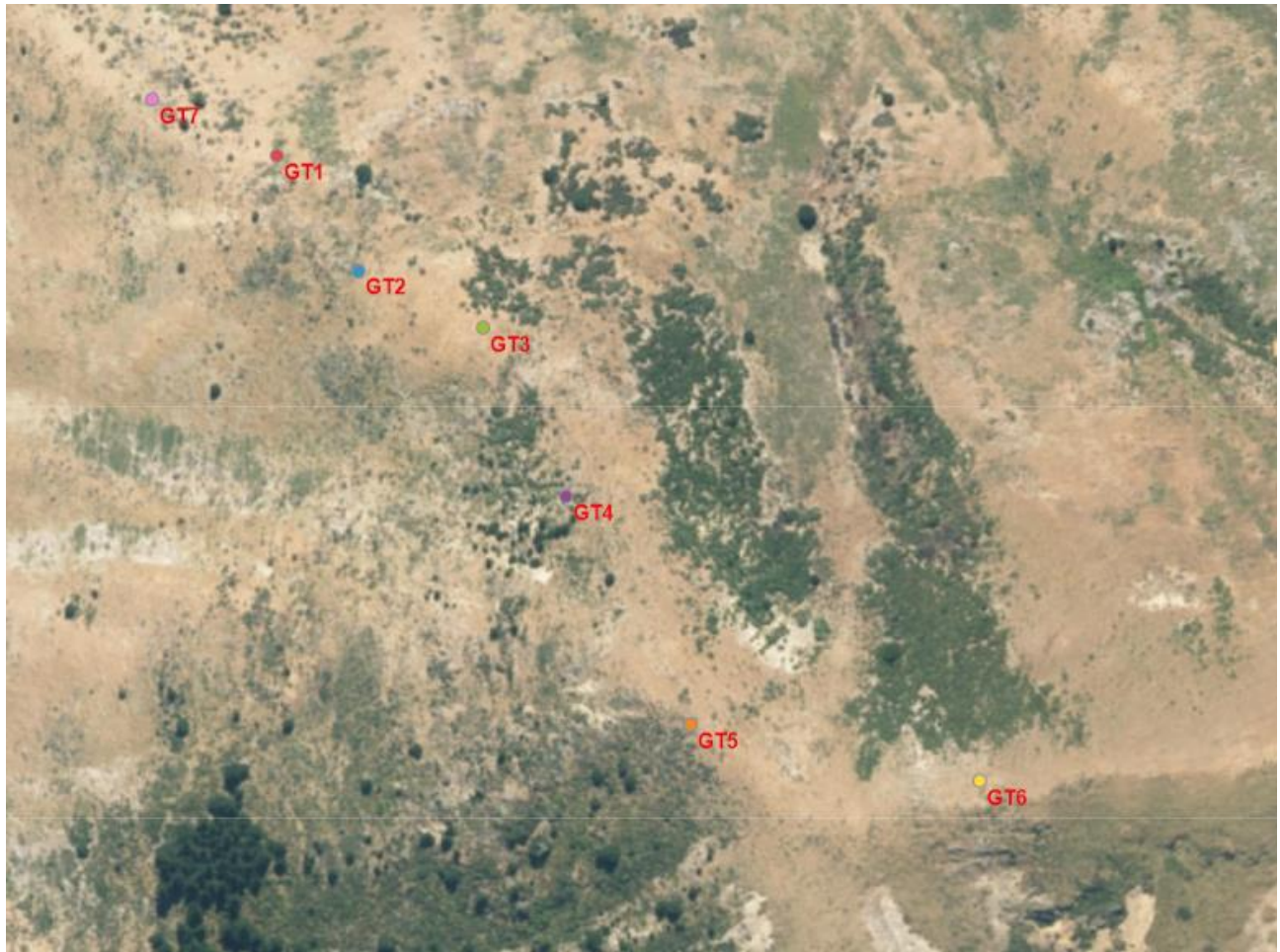


Figure 2: Map of the field site. GT6 is the highest elevation, GT7 is the lowest elevation. All other units follow a ridgeline decreasing in elevation from GT6 to GT7.

Name	Lat	Long	Description of outcrop
GT1	-42.77805556	172.5627778	Strange breccia (agglomerate outcrop)
GT2	-42.77861111	172.5633333	Small (~1m) outcrop on ridge at change from Matagouri to Grass. Dark grey to black rock - sugary appearance.
GT3	-42.77888889	172.5641667	Large (~5m across) but sporadic outcrop of brown to dark grey rock. Has small scree deposits in front of outcrop - very similar to Torlesse scree
GT4	-42.77972222	172.5647222	Small (0.5m) outcrop, dark grey. Unsure lithology - gives off distinctive 'silica' smell when hammered
GT5	-42.78083333	172.5655556	Outcrop at top of ridge - coarse grained igneous. Most of way from GT4 to GT5 is siliceous until GT5, which is coarse igneous
GT6	-42.78111111	172.5675	Two outcrops near saddle to larger hill. Lighter rock is porphyritic microsyenite, other is fine grained and dark. Unsure of lithology.
GT7	-42.77777778	172.5619444	Coarse grained igneous below breccia outcrop. Location is below gnarly tree on ridgeline



Figure 3a: GT1 - Agglomerate outcrop on Syenite hill.



Figure 3b: Close up of GT1

Major Element Composition of GT1

Matrix vs. Clasts in Matrix

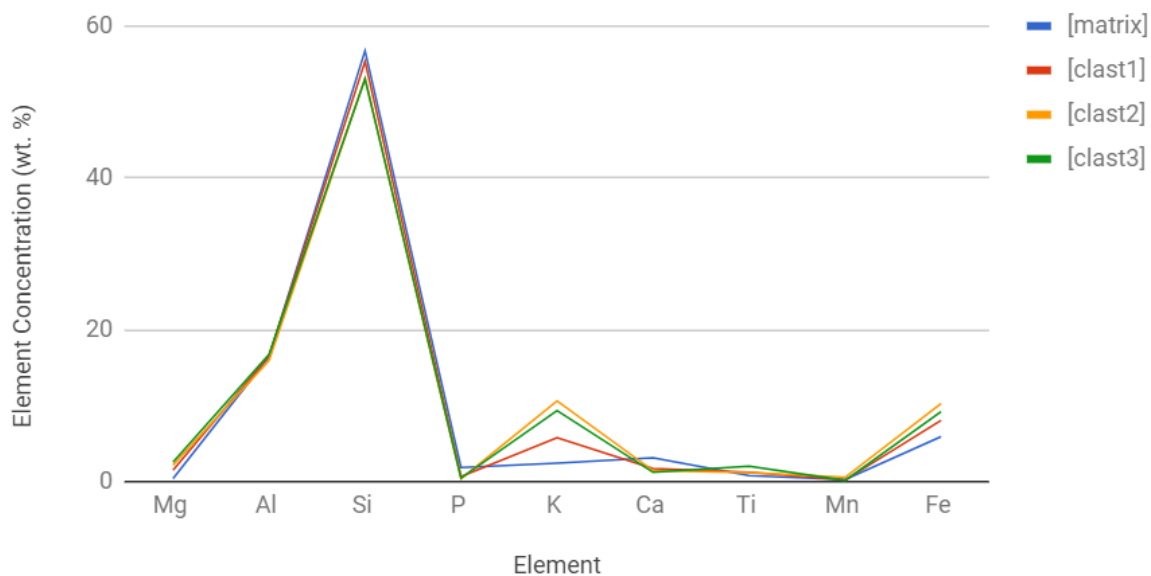


Figure 4: Elemental Composition of major elements in GT1 matrix vs. three individual clasts in the matrix.

Minor Element Composition of GT1

Matrix vs. Clasts in Matrix

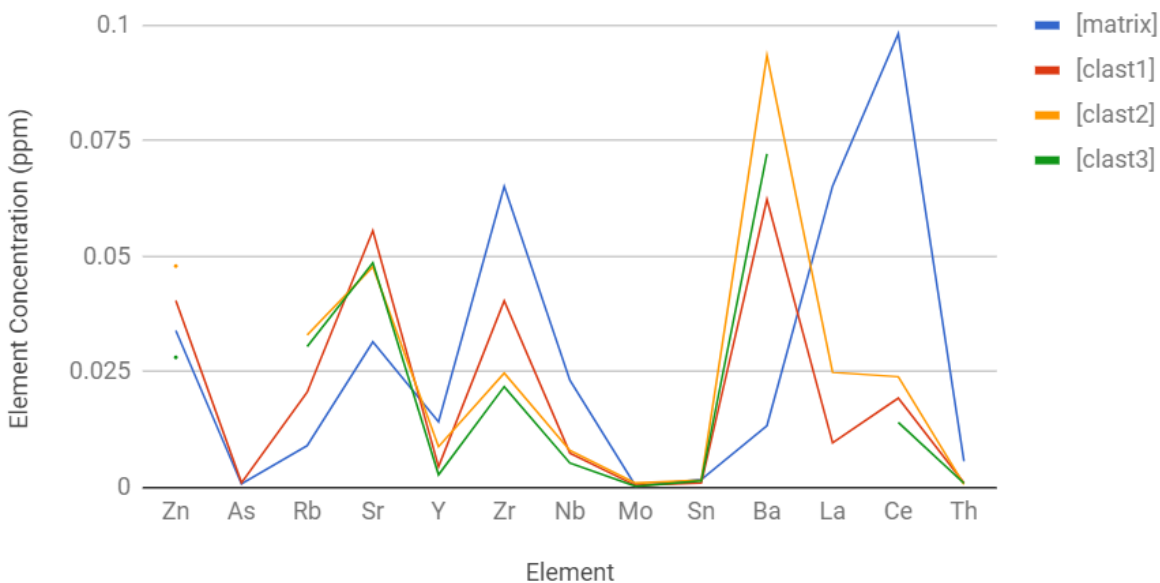


Figure 5: Elemental Composition of minor elements in GT1 matrix vs individual clasts cemented in the matrix.

Major Element Composition of GT1 vs. Adjacent Outcrop GT7

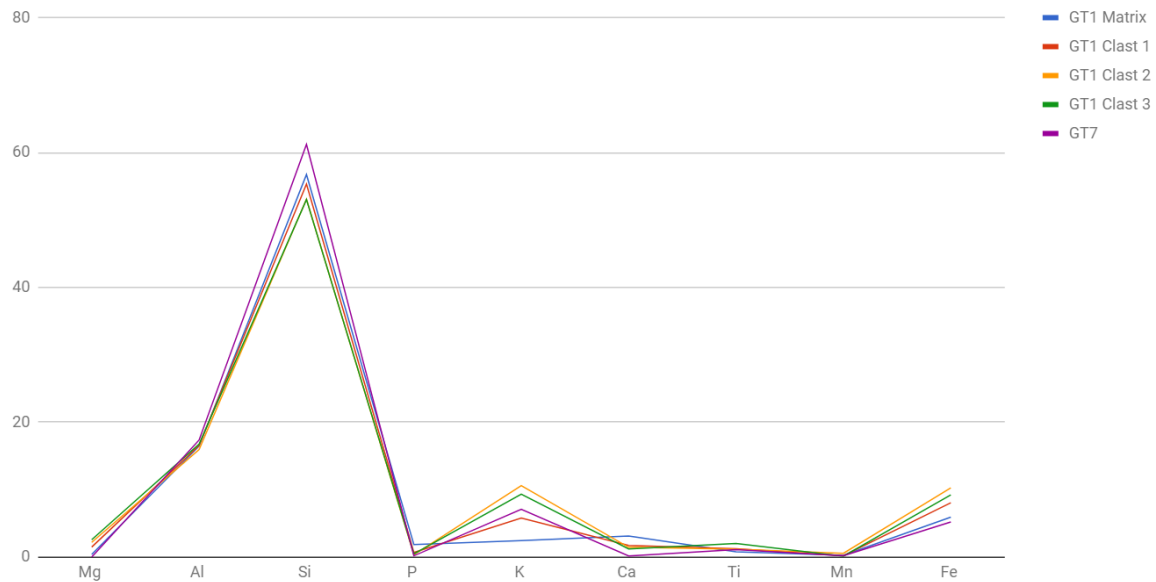


Figure 6: Elemental Composition of major elements in GT1 matrix and individual clasts cemented in the matrix, compared with an adjacent outcrop GT7.

Major Element Composition of Outcrops on Hill

Elemental Percentage by Outcrop Altitude

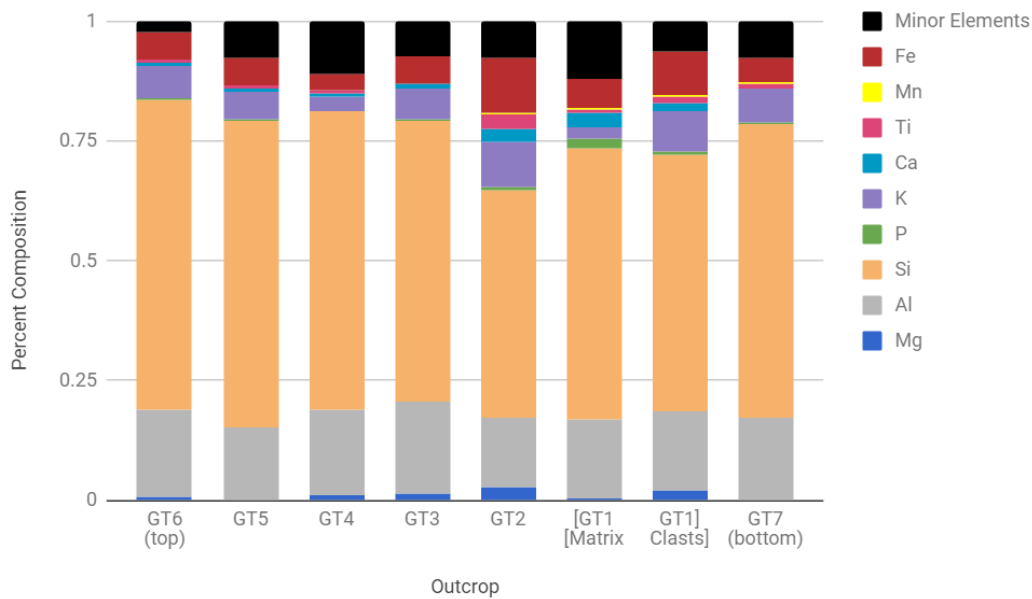


Figure 7: Elemental Composition of major elements in GT1 through GT7. Outcrop altitude decreases to the right (i.e. GT6 is at the highest elevation, GT7 is at the lowest elevation). The black at the top of each column incorporates all minor elements, and any elemental composition that was not read by the pXRF.



Figure 8: Image of weathered plutonic unloading sheets, taken facing toward the side of Diorite Hill near the outcrop GT5.

Discussion and Conclusions

From Figure 4 it can be seen that the major element compositions in the clasts vs the matrix are roughly in line with each other; K and Fe show slight differences among the samples. Figure 5 shows wider variation in minor elements for the same comparison, with the most significant composition differences between Ce, Ba, Zr, and Sr respectively.

The lack of K, Ba, and Sr, and the abundance of Ce, and Zr in the matrix when compared with the clasts is significant, and suggests that the matrix and clasts are different rocks. The most likely event to have caused this formation appears to be the outward escape of magma from an ancient melt, whereby a narrow tunnel of molten rock trapped fragments of cooled crustal material on its way. As there is not significant difference in most major elements, it is safe to say that the matrix and clasts are comprised highly of the same minerals with some variation. Hydrothermal alteration, and difference in cooling rates between the matrix and clasts could both contribute to this variation.

This hypothesis makes sense in light of the perceived ancient formative environment, as the hill behind the field site is a massive diorite pluton. Weaver and Pankhurst help reveal this in their map of this area; one landscape feature in particular also makes this fairly noticeable (see Figure 8). What appear to be large unloading sheets are pictured on the left; vertical lines show the up-down orientation of these weathered sheets. The diorite dome sits within a mountainous framework of syenite, where are all of the currently presented outcrops are located. If pluton formation was a driving force in this area when these rocks were formed, it is quite possible that a hot chunk of magma escaped to the cooler, outer edge of the melt where a crust had already developed. The slow cooling of this crust would explain the visible crystals in the clasts, while the invisible crystals of the matrix suggest rapid cooling. The hypothesis supports rapid cooling, as escapement of magma to the surface would be a natural cause.

It is entirely possible that Weaver and Pankhurst glossed over the outcrops I investigated, as they comprise a very small percentage of an area that is mostly syenite. This outcrop tells an interesting story about the formative environment of these rocks, however, and provides insight

as to magma escape and clast entrapment causes variations in the elements found between clasts and matrix cement.

Reference Work

S. D. Weaver & R.J. Pankhurst (2010) A precise Rb- Sr age for the Mandamus Igneous Complex, North Canterbury, and regional tectonic implications, New Zealand Journal of Geology and Geophysics, 34:3, 341-345, DOI: 10.1080/00288306.1991.9514472